

Risk Assessment and Mitigation

Cohort 1 Group 7

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Risk Management Process and Risk Register Format

1. Risk Management Process

The team strictly followed structured risk management processes as outlined in the lecture throughout the project. As project and team were small, the process was kept simple but consistent. Via team meeting sessions, potential risks were addressed and discussed early to identify and mitigate it efficiently.

- ❖ Risk Identification

The known and unknown risks were discussed and identified early in the project during brainstorming session. To make identification easier risks were categorized in four categories (Technology, People, Requirements, Estimation).

- ❖ Risk Analysis

Each risk identified was rated for likelihood and severity on a Low–Medium–High scale. Risks with both high likelihood and severity were taken utmost caution while both low likelihood and severity were eliminated.

- ❖ Risk Mitigation

Preventive and response plan was made for every identified risk. Prevention plan reduced the likelihood while response plan smoothly solved the risks.

- ❖ Risk Monitoring

For every identified risk, risks were assigned to the owner responsible for monitoring and updating each progress. The risk register was updated upon each encounter with unknown risks and risk was resolved.

2. Format of the Risk Register

❖ ID	Risk Identifier (R1, R2 ...)
❖ Category	Risk category (Technology, People, Requirements, Estimation)
❖ Description	Brief description of the potential risk
❖ Likelihood	Probability of occurrence (Low, Medium, High)
❖ Severity	Impact it causes on the project (Low, Medium, High)
❖ Mitigation	Preventive and response plan
❖ Owner	Person tasked with monitoring and managing the risk

3. Justification of the Process

The process was simple yet effective and required collaboration among all team members, making risk prevention more precise and on time. The improvement was continuously made as project progressed.

Risk Register

ID	Type	Description	Severity	Likelihood	Mitigation	Owner
R1	Technology	Bugs or Crashes causing delays	L	H	Fixing when arise	Matt, William
R2	Technology	Low quality code or unreadable code affecting maintainability	L	H	Following coding standard	William, Matt
R3	Technology	Lose project files	H	L	Frequent Backup and Version control	All
R4	Technology	Build or executable not working as expected	H	M	Debug and rebuild it	Matt, William
R5	Technology	Poor project organization or traceability	M	M	Consistent documentation and naming	All
R6	People	Key member leaves or unavailable	H	L	Distribute tasks evenly and document regularly to smooth overtake	All
R7	People	Miscommunication leads to wrong implementation and low performance	H	M	Regular Team Meeting to verify and check other's progress	All
R8	Requirements	Unclear or changing requirements lead to rework	M	M	Regular client meeting	Max
R9	People	Team conflict	M	L	Team building events and encourage respectful discussion	All
R10	Requirements	Misinterpretation of requirements	H	M	Team Meeting to requirement interpretation	All

R11	Requirements	Accessibility limitation not counted	M	L	Accessibility menu for adjustments	Ronny, Bailey
R12	Requirements	Inconsistent Art Style	M	M	Team poll (acceptable/not)	Bailey, Ronny
R13	Requirements	Storyline lack clarity	M	M	Team plotting	Max
R14	Requirements	Poor user interface	H	M	Testing	Bailey, Ronny
R15	Estimation	Wrong time distribution	H	M	Use Gantt Chart and break tasks smaller	Enkhee, Kyle
R16	Estimation	Unbalanced game difficulty (too easy or too difficult)	M	M	Refine balance through tuning, testing, and feedback.	Bailey, Ronny
R17	Estimation	Design is confusing and not engaging (Map)	H	M	Design review session	Ronny, Bailey
R18	Technology	Game is sluggish due to code or assets	H	L	Optimize inefficient code loops, reduce asset sizes and complexity	William, Ronny
R19	Requirements	Violation of existing intellectual properties (licenses etc.)	H	M	Verify asset sources	All
R20	Requirements	Make the link public, resulting in loss of project data	H	L	Keep repositories private	Ronny
R21	Requirements	Not mentioning used assets and licenses	M	L	Mention it in justification	All
R22	Requirements	Lack of justification in the documentation	M	L	Include referencing and enough justification	All

R23	Technology	Use of AI	H	H	only for graphical assets (like character/building assets) or sounds	All
R24	Requirements	Assets misuse	M	L	Follow guidelines and properly distribute it	William, Matt
R25	Requirements	Badly planned project structure	M	M	Clear hierarchy and structure	Enkhee, Kyle